

The risk posed by *Aedes aegypti* in Cyprus and the wider Mediterranean region



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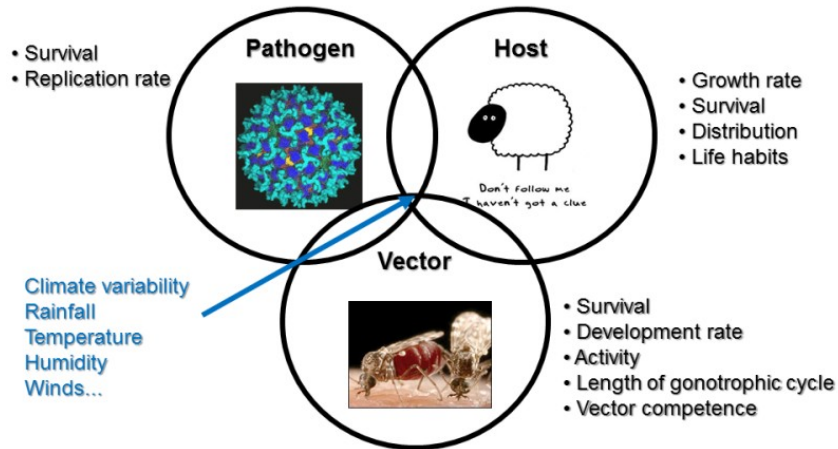
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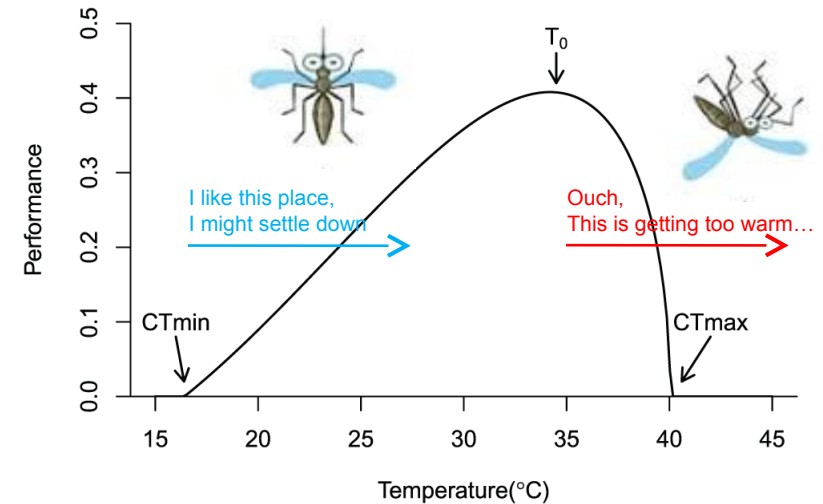
Climate & Vector-borne diseases (VBDs)

VBDs are climate sensitive

Diseases transmitted by blood sucking arthropods



Vectorial capacity = $F(T^\circ)$



Lafferty KD and Mordecai EA 2016 - [F1000Research 2016, 5:2040](#)

Modelling the impact of climate variability on VBD burden, development of early warning systems (seasonal to climate change time scales).

Impact of VBD



2nd Plague pandemic
14th century



500k deaths/
year in Africa

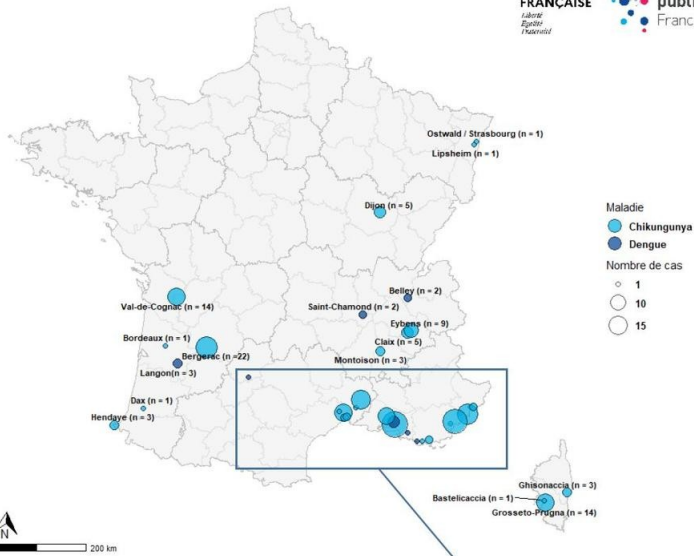


Zika outbreak, Latin America 2015-16



Yellow Fever in Angola-DRC 2015-16

Localisation des épisodes de transmission autochtone - 2025



Distribution of CHIKV & ZIKV autochthonous events in France in 2025



Bluetongue in Europe, 2006



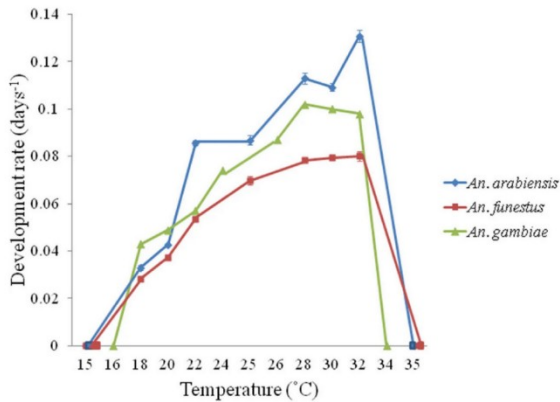
Xylella fastidiosa in Italy

- Large historical impacts of VBDs
- Autochthonous transmission of arboviruses has increased in France & Italy

Epidemiological parameters affected by T°

Development & mortality rate $\mu(T^\circ)$

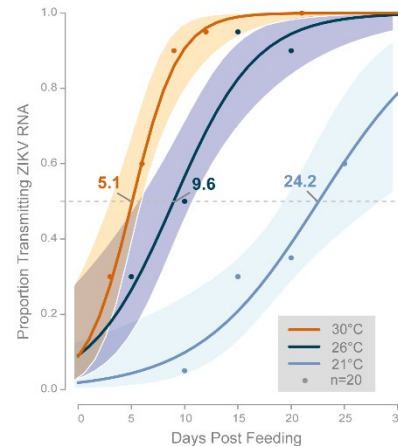
Example for *Anopheles malaria* vectors



Lyons et al. 2013. [Parasites Vectors](#) 6, 104

Extrinsic Incubation period $EIP(T^\circ)$

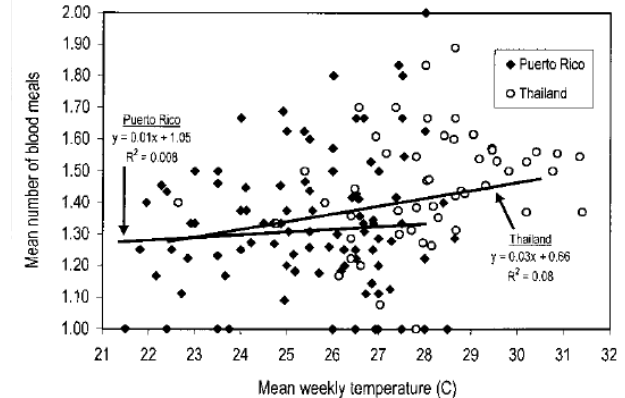
Example for *Ae. aegypti* and Zika virus



Winokur et al. 2020. [PLoS Negl Trop Dis](#) 14(3): e0008047

Biting rate $a(T^\circ)$

Example for *Ae. aegypti* for two sites in Puerto Rico and Thailand

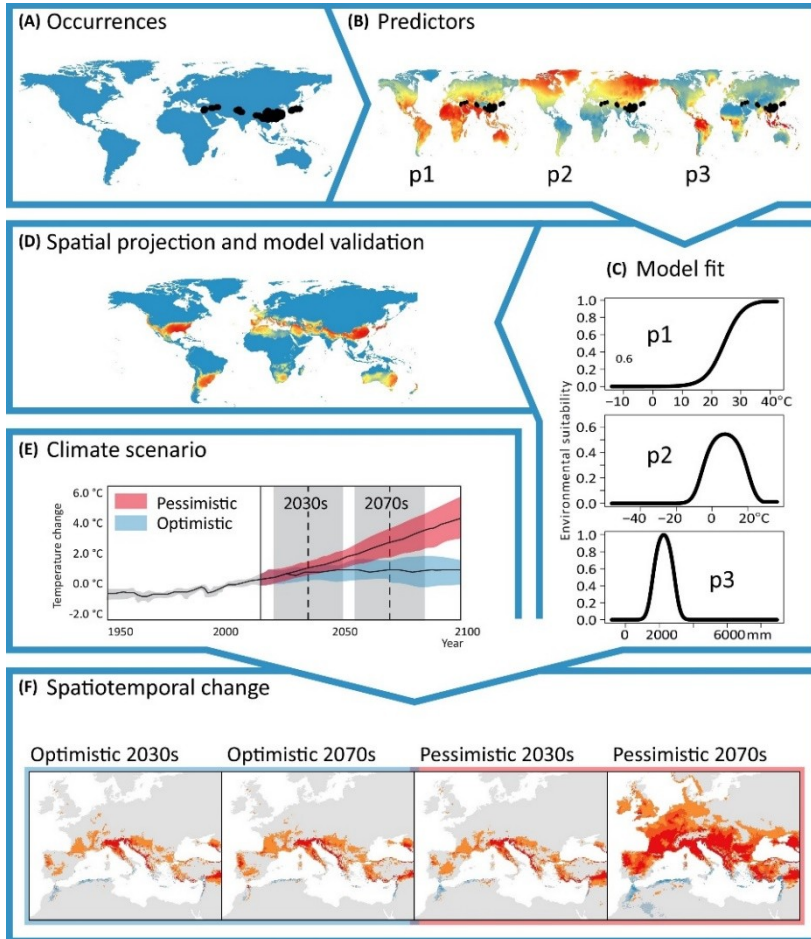


Scott et al. 2000. [J Med Entomol](#). 37(1): 89-101

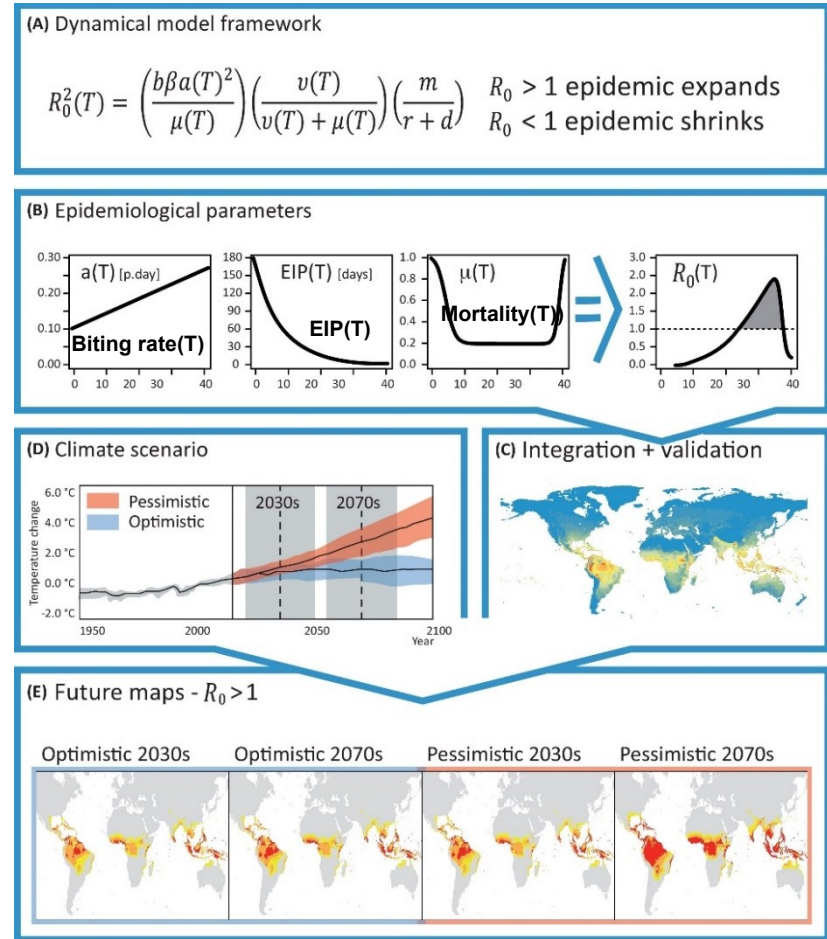
- **Temperature impacts** the biting rate $a(T^\circ)$, development rate $\mu(T^\circ)$, and the extrinsic incubation period $EIP(T^\circ)$ that is the time required for the vector to become infectious following an infected bloodmeal.
- **Rainfall** provides breeding sites. Extreme rainfall can lead to flushing of larvae.
- Other factors matter (land use, travel and trade, socio-economic factors, human behaviour, shape of health services...).

Methods to model risk

Statistical models



Mechanistic models

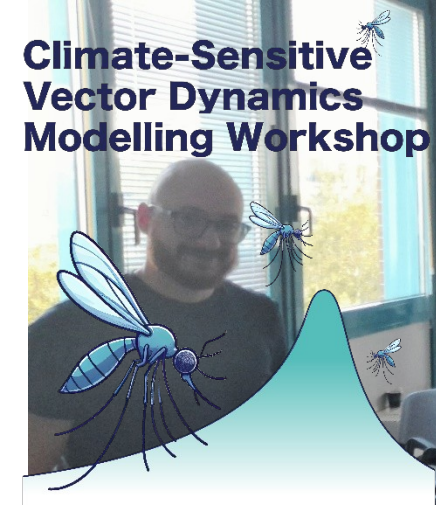


Stat models: Maxent, BRTs, Bayesian models, Mahalanobis distance...

Mechanistic models: SEIR/SIR, R_0 , Fuzzy logic, climate envelope...

Fig. 1 & 2 Tjaden et al. (2018). *Trends in Parasitology* 34(3): 227-245. <http://dx.doi.org/10.1016/j.pt.2017.11.006>

1st Vector modelling workshop in Bologna, 19-20 Sep 2024



Bologna, 19-20 September 2024



<https://www.vectormodelling.com/Bologna2024/>

Key facts about *Aedes aegypti*

AÈDES AEGYPTI

THE YELLOW FEVER MOSQUITO:
ITS LIFE HISTORY, BIONOMICS
AND STRUCTURE



BY

SIR RICKARD CHRISTOPHERS

CAMBRIDGE UNIVERSITY PRESS

R. Christophers. (1960) *Aedes Aegypti* (L.) The Yellow Fever Mosquito: Its Life History, Bionomics and Structure. 1st Edition. ISBN-13: 978-0521113021

- *Ae. aegypti* is competent to transmit Yellow fever, dengue, Zika and chikungunya
- It feeds almost exclusively on humans
- It uses man-made water containers to lay eggs and is well adapted to the urban setting
- 40% of the world population is at risk of contracting dengue fever
- It evolved from a sylvatic form (*Ae. aegypti formosus*) to a domestic form *Aedes aegypti aegypti* and originates from Africa – original lineage from Madagascar 7My ago

Historical distribution of *Aedes aegypti*

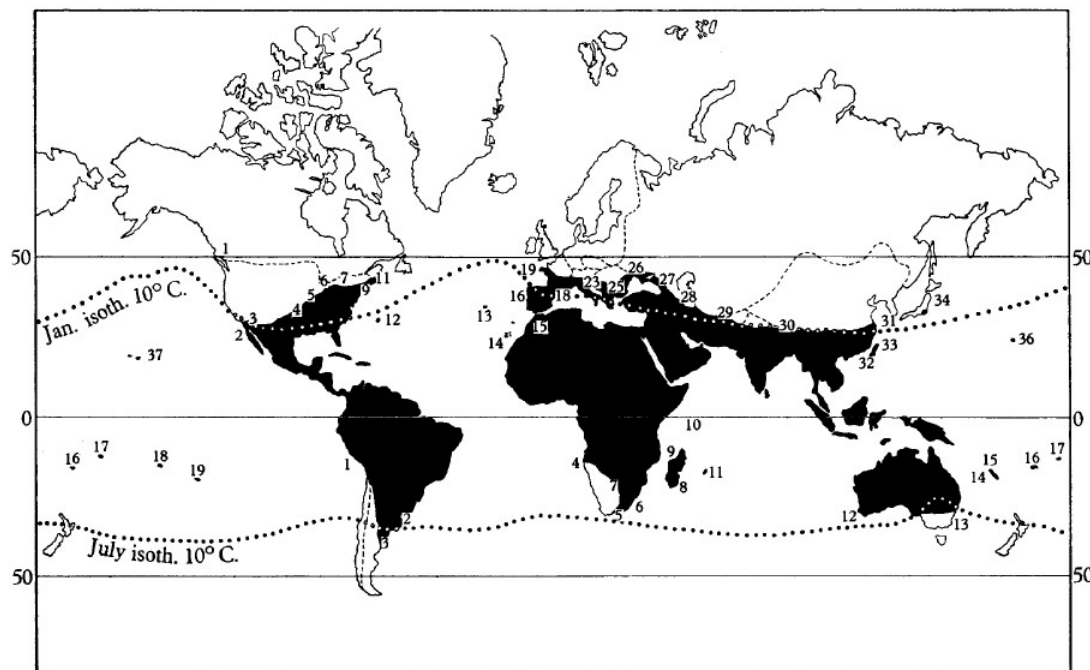


Figure 1. Map showing limits of distribution.

Map of the world showing northern and southern limits of the recorded distribution of *Aedes aegypti* and isotherms of 10° C.

The numbers are those given to recorded localities as set out in Tables 1 and 2.

Figures in brackets in the tables give the numbers in the list of references to geographical distribution on pp. 50–2.

Table 1. Recorded northern limits of distribution.*

North America

- 1 British Columbia (37), 50° N.†
- 2 Lower California (62), 32° N.
- 3 Arizona (9), 32° N.
- 4 Oklahoma (57), 36° N.‡
- 5 Missouri (40, 62), 38° N.
- 6 Indiana (40, 62), 40° N.
- 7 Virginia (40, 62), 38° N.
- 8 Maryland (40, 62), 39° N.
- 9 Philadelphia (16), 40° N.†
- 10 New York (16), 40° 52' N.†
- 11 Boston (16), 42° 27' N.†

Atlantic Islands

- 12 Bermuda (6), 32° 20' N.
- 13 Azores (62), 38° 35' N.
- 14 Canary Islands (18, 47), 28° 40' N.
- 15 Morocco (Tangier) (17), 35° 42' N.

Europe

- 16 Portugal (Badajos) (35), 38° N.
- 17 Spain (north-west point) (13), 43° N.
- 18 Spain (Ebro) (27), 41° N.

Europe (cont.)

- 19 France (Brest) (42), 48° 24' N.
- 20 France (Dol) (42, 62), 48° 31' N.
- 21 France (Bordeaux) (59), 44° 48' N.
- 22 Italy (Genoa) (42), 44° 25' N.
- 23 Italy (Ravenna) (42), 44° 25' N.
- 24 Bosnia (3), 44° N.
- 25 Macedonia (42), 41° N.
- 26 Russia (Odessa) (41), 46° 30' N.‡
- 27 Russia (Sukhum) (58), 43° N.
- 28 Russia (Baku) (1), 40° 25' N.

Asia

- 29 India (Peshawar) (8), 34° N.
- 30 India (Assam) (8), 26° N.
- 31 China (Shanghai) (29), 31° 10' N.
- 32 China (Amoy) (30), 24° 40' N.
- 33 Formosa (29, 42), 24° N.
- 34 Japan (Tokyo) (42, 62), 35° 40' N.
- 35 Japan (Okayama) (42), 34° 22' N.

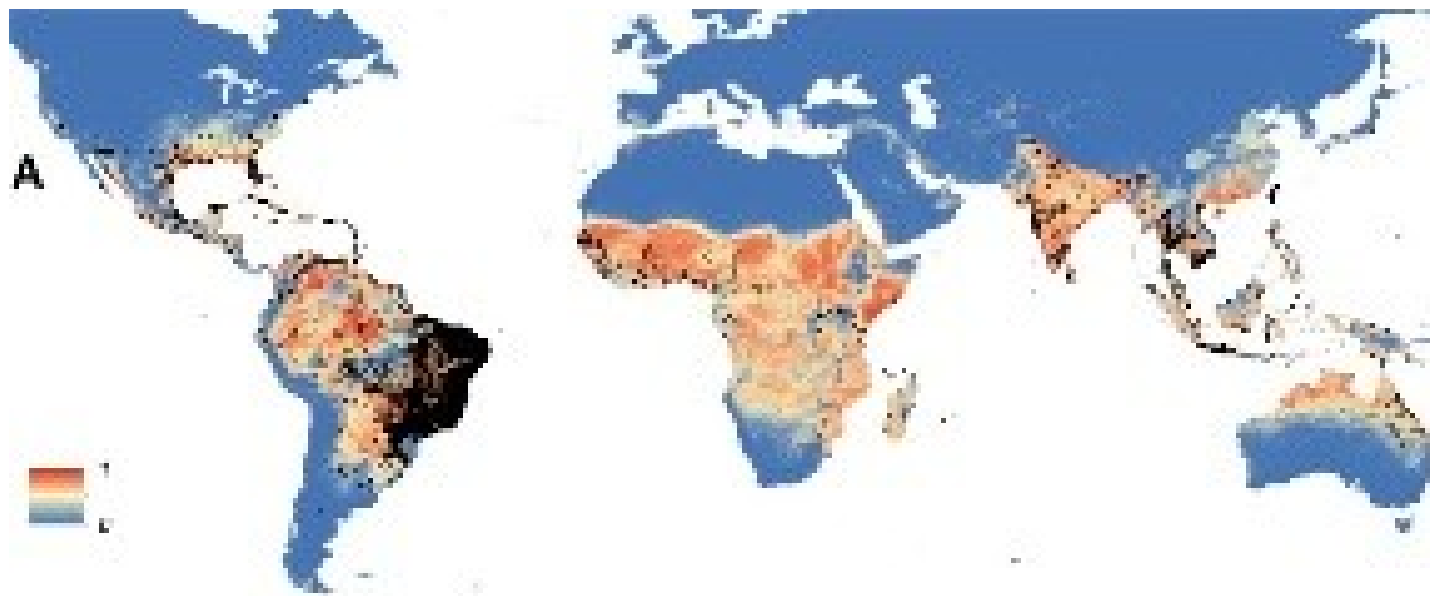
North Pacific Islands

- 36 Wake Island (56), 20° N.
- 37 Hawaii (11, 39), 20° N.

R. Christophers. (1960) Aedes Aegypti (L.) The Yellow Fever Mosquito: Its Life History, Bionomics and Structure. 1st Edition. ISBN-13: 978-0521113021

Distribution of *Ae. aegypti* before control measures were undertaken post WWII.
Southernmost occurrence: Buenos Aires; northernmost: British Colombia.
Ae. aegypti traveled by boat and was found in major sea ports...

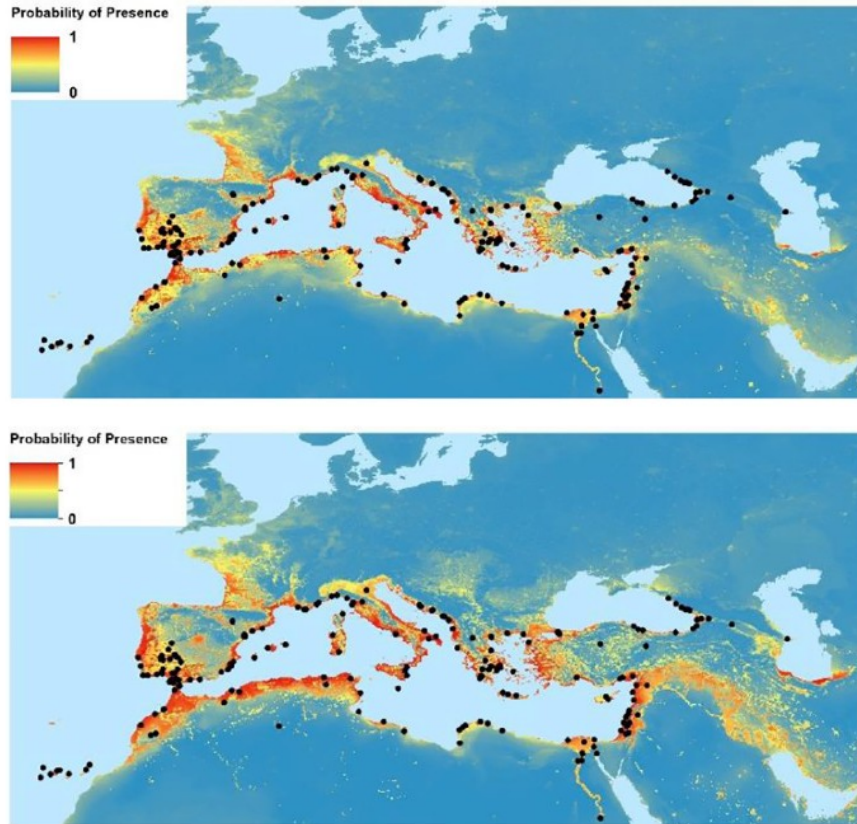
Distribution of *Aedes aegypti* in 2014



Kraemer et al. (2015) The global distribution of the arbovirus vectors *Aedes aegypti* and *Ae. albopictus* [eLife 4:e08347](https://doi.org/10.1371/journal.pone.0135933).

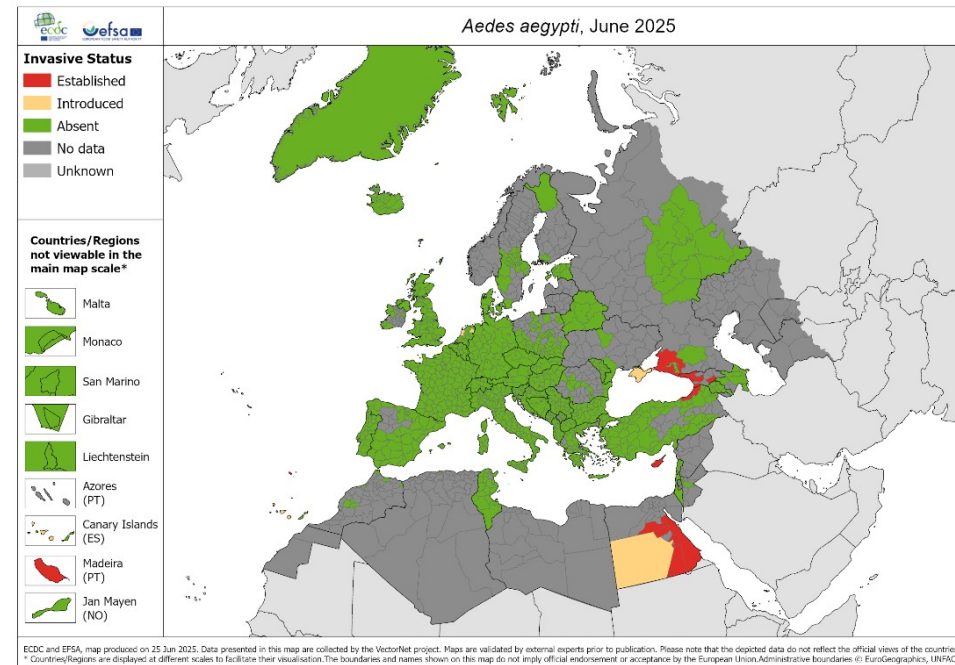
Ae. aegypti has a tropical/semi tropical distribution and is abundant in Latin America, the southern US states, India, Southeast Asia, Australia, and Africa.

Aedes aegypti in Europe and the Mediterranean region



Western Palearctic region historical 1910 (top) and 2015 (bottom) modelled suitability for *Aedes aegypti* with population in the covariates. Black dots: recorded historical (up to 1955) presence points.

Wint et al. (2022). *Sci Total Environ.*, 847:157566. <https://doi.org/10.1016/j.scitotenv.2022.157566>



Observed distribution of *Ae. aegypti* in Europe in June 2025

European Centre for Disease Prevention and Control and European Food Safety Authority. Mosquito maps [internet]. Stockholm: ECDC; 2025. Available from: <https://ecdc.europa.eu/en/disease-vectors/surveillance-and-disease-data/mosquito-maps>

Ae. aegypti was present over southern Europe before control measures were undertaken. It is now present over the eastern coasts of the Black sea, northwestern Turkey and **Cyprus**.



A yellow quarantine flag, signaling yellow fever, is raised on a ship anchored at sea some distance from a port in this watercolor by E. Schwarz-Lenoir painted some time between 1920 and 1950. CC-BY 4.0 via the Wellcome Collection.

<https://www.thinkglobalhealth.org/article/plague-ships-shotgun-mobs-and-saffron-scurge>

Ae. aegypti modeling work in Europe and Cyprus

Data & Methods

Data:

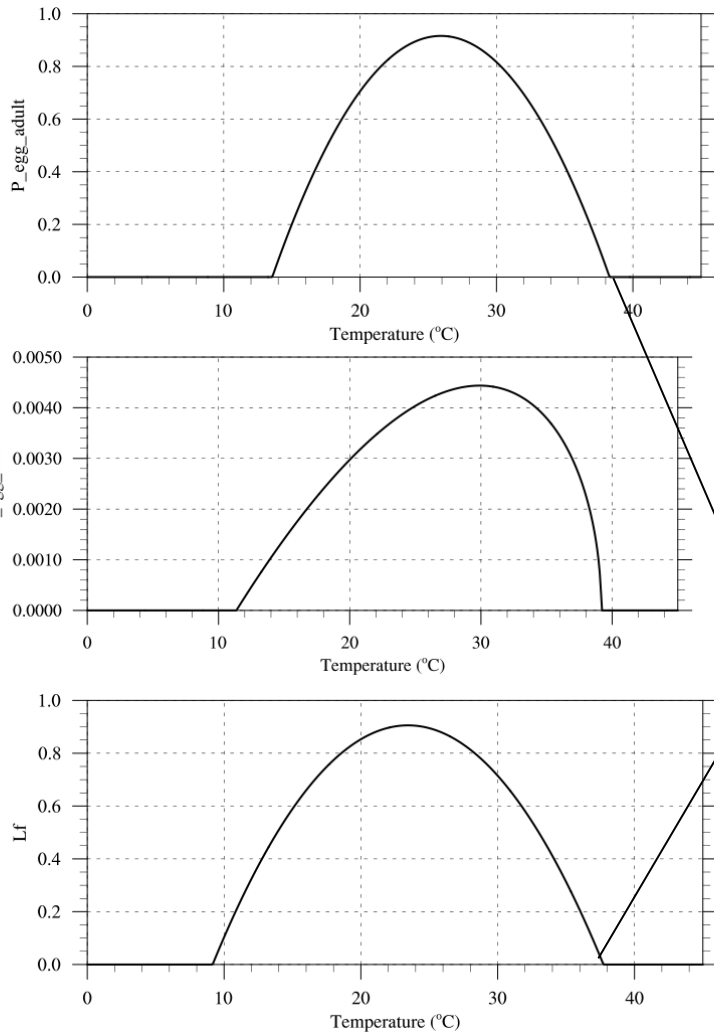
- *Ae. aegypti* historical occurrence data (black dots) derived from [Wint et al., 2022](#)
- *Ae. aegypti* ovitrap data for Larnaca, Cyprus (Dr. Marlen Vasquez, Cyprus University of Technology, Limassol) [<https://doi.org/10.1101/2024.12.17.628941>]
- Lower resolution 25km input climate data derived from EOBS data for Europe [<https://doi.org/10.24381/cds.151d3ec6>]
- High resolution 1km input climate data (rainfall & temperature) derived from EMO-1 data for Cyprus [<https://doi.org/10.2905/0BD84BE4-CEC8-4180-97A6-8B3ADAAAC4D26>]

Ae. aegypti models:

- **Model 1:** Simulated probability of occurrence from [Kraemer et al., 2015](#)
- **Model 2:** Mean annual temperature over 10°C (“old” WHO model & Christophers, 1960)
- **Model 3:** Oviposition rate derived from Yang et al. 2009 [<https://doi.org/10.1017/S0950268809002040>]
- **Model 4:** VECTRI albopictus model ([Garrido Zornoza et al., 2024](#)) with modified adult and larval mortality schemes for *Ae. aegypti* based on Caldwell et al. 2021.

Analysis for Cyprus is mainly based on the *Ae. aegypti* version of the VECTRI model (**Model 4**)

VECTRI *Ae. albopictus* -> *aegypti* prototype



JOURNAL OF THE ROYAL SOCIETY INTERFACE

Garrido Zornoza M., et al. (2024). J. R. Soc. Interface 21:20240319

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The effect of climate change and temperature extremes on *Aedes albopictus* populations: a regional case study for Italy

Miguel Garrido Zornoza, Cyril Caminade and Adrian M. Tompkins

Published: 06 November 2024 <https://doi.org/10.1098/rsif.2024.0319>

Details

References

Related

Figures

This Issue

Table 3 Fitted thermal responses for *Ae. aegypti* life-history traits.

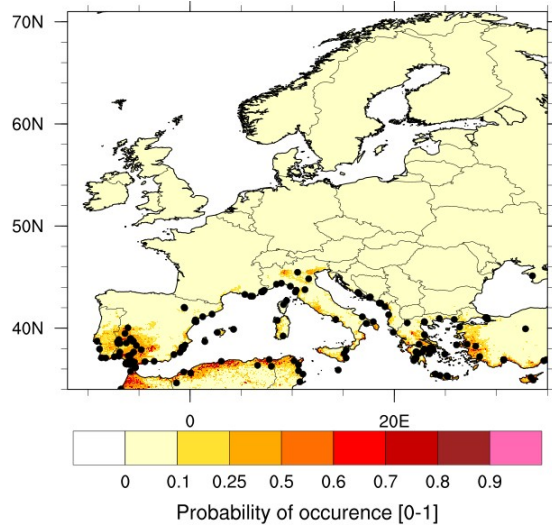
From: [Climate predicts geographic and temporal variation in mosquito-borne disease dynamics on two continents](#)

Trait	Definition	Function	c	T_0	T_m
a	Biting rate (day^{-1})	Brière	2.02×10^{-4}	13.35	40.08
EFD	Eggs laid per female per day	Brière	8.56×10^{-3}	14.58	34.61
pEA	Probability of mosquito egg-to-adult survival	Quadratic	-5.99×10^{-3}	13.56	38.29
MDR	Mosquito egg-to-adult development rate (day^{-1})	Brière	7.86×10^{-54}	11.36	39.17
Lf	Adult mosquito lifespan (days)	Quadratic	-1.48×10^{-1}	9.16	37.73
b	Probability of mosquito infectiousness	Brière	8.49×10^{-4}	17.05	35.83
pMI	Probability of mosquito infection	Brière	4.91×10^{-4}	12.22	37.46
PDR	Parasite development rate (day^{-1})	Brière	6.56×10^{-5}	10.68	45.90

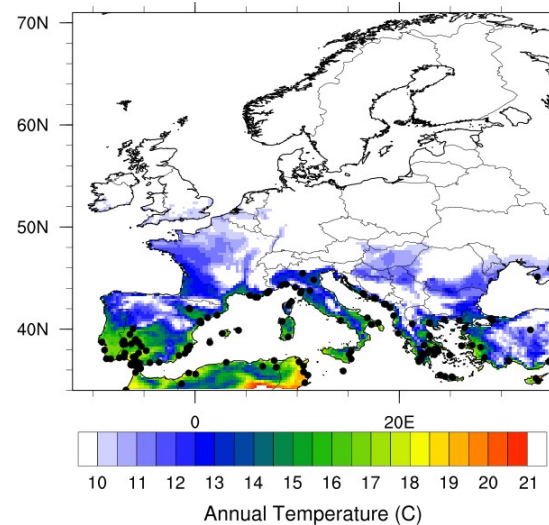
Traits were fit to a Brière $[cT(T - T_0)(T_m - T)^{\frac{1}{2}}]$ or a quadratic $[c(T - T_m)(T - T_0)]$ function where T represents temperature. T_0 and T_m are the critical thermal minimum and maximum, respectively, and c is the rate constant. Thermal responses were fit by ref. [19](#) and also used in ref. [30](#). Parasite development rate was measured as the virus extrinsic incubation rate.

Caldwell. et al. (2021). Nat Commun 12, 1233.

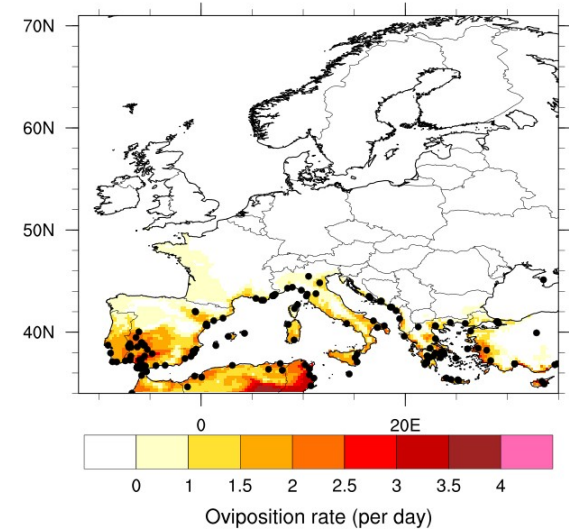
Comparison of *Ae. aegypti* models over Europe



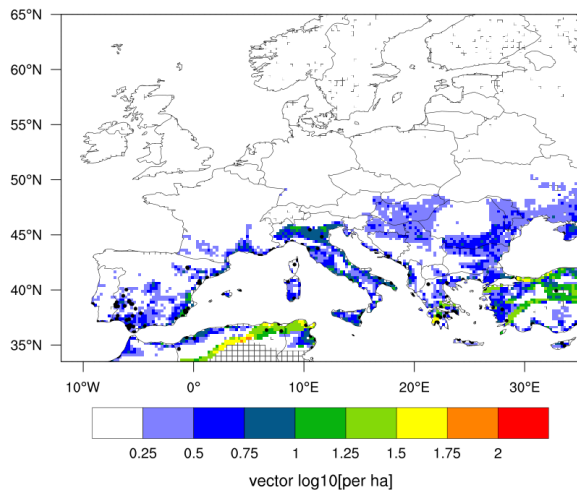
Model 1 - Kraemer et al., 2015



Model 2 – Tann > 10°C



Model 3 - Oviposition(T°)
Yang et al. 2009



Model 4 – *Ae. aegypti* version of VECTRI (simulated adult density)

Distribution of Ae. aegypti in Europe based on different models. Rainfall and temperature data was derived from the EOBS dataset (0.25°) for 2011-2021.

Models based on different assumptions confirm that southern Europe is still climatically suitable for the establishment of *Ae. aegypti*.

Ae. aegypti in Cyprus

New Results

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doi: <https://doi.org/10.1101/2024.12.17.628941>

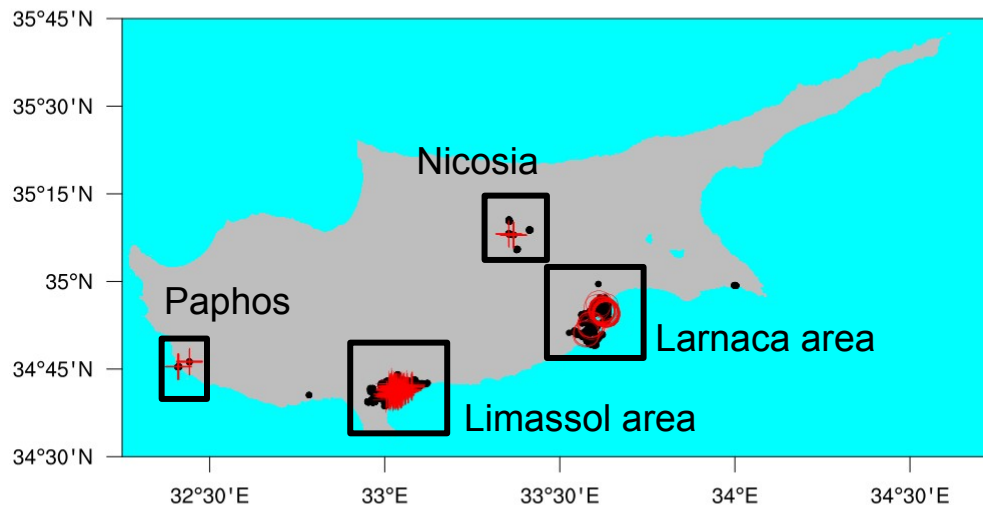
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Subject Area

Ecology

Abstract Full Text Info/History Metrics

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Ovitrap data collected between Oct 2022 and June 2023 over Cyprus (CIT team led by M. Vasquez & IAEA team led by J. Bouyer).

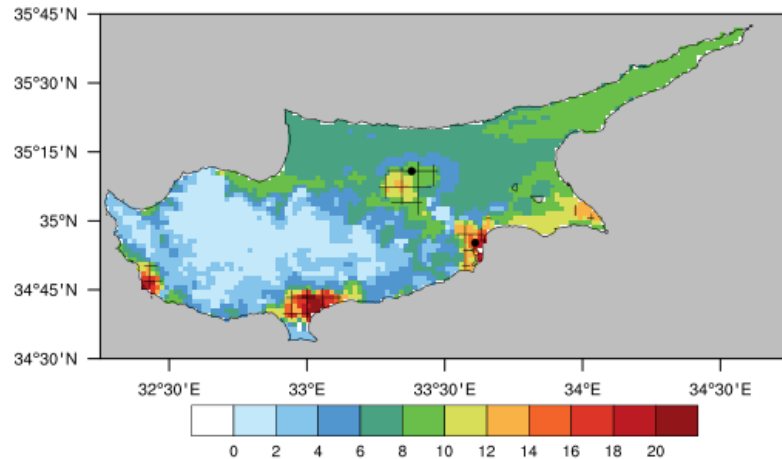
Negative traps are depicted by black dots, positive *Ae. aegypti* traps are denoted by red circles and positive *Ae. albopictus* traps are indicated by red crosses.

Aedes mosquitoes are present in urban areas of Cyprus.

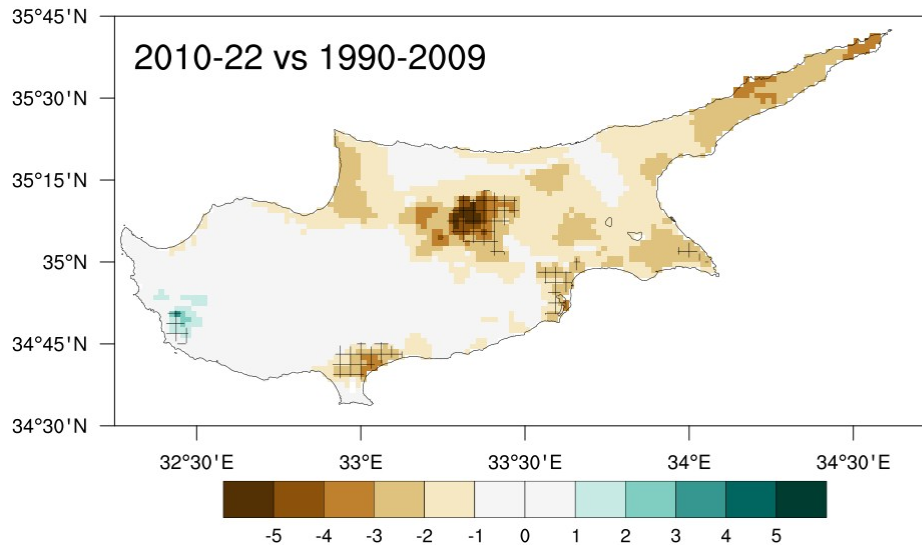
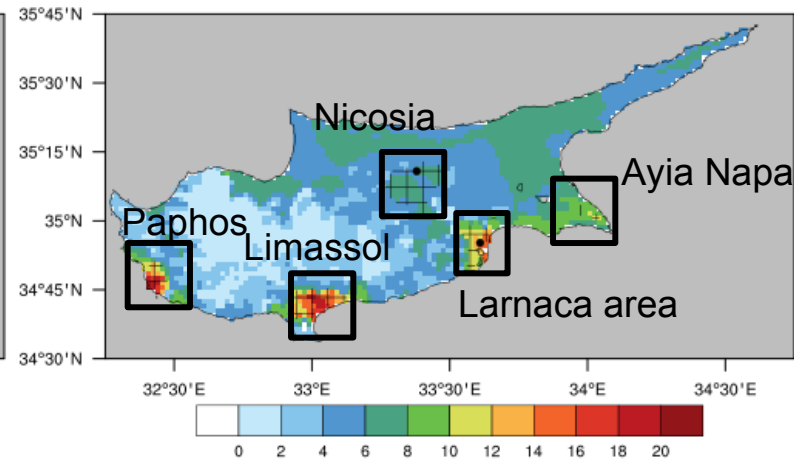
Caminade et al. (2024)
<https://doi.org/10.1101/2024.12.17.628941>

Simulated distribution of *Ae. aegypti* and changes in Cyprus

1990-2009



2010-2022

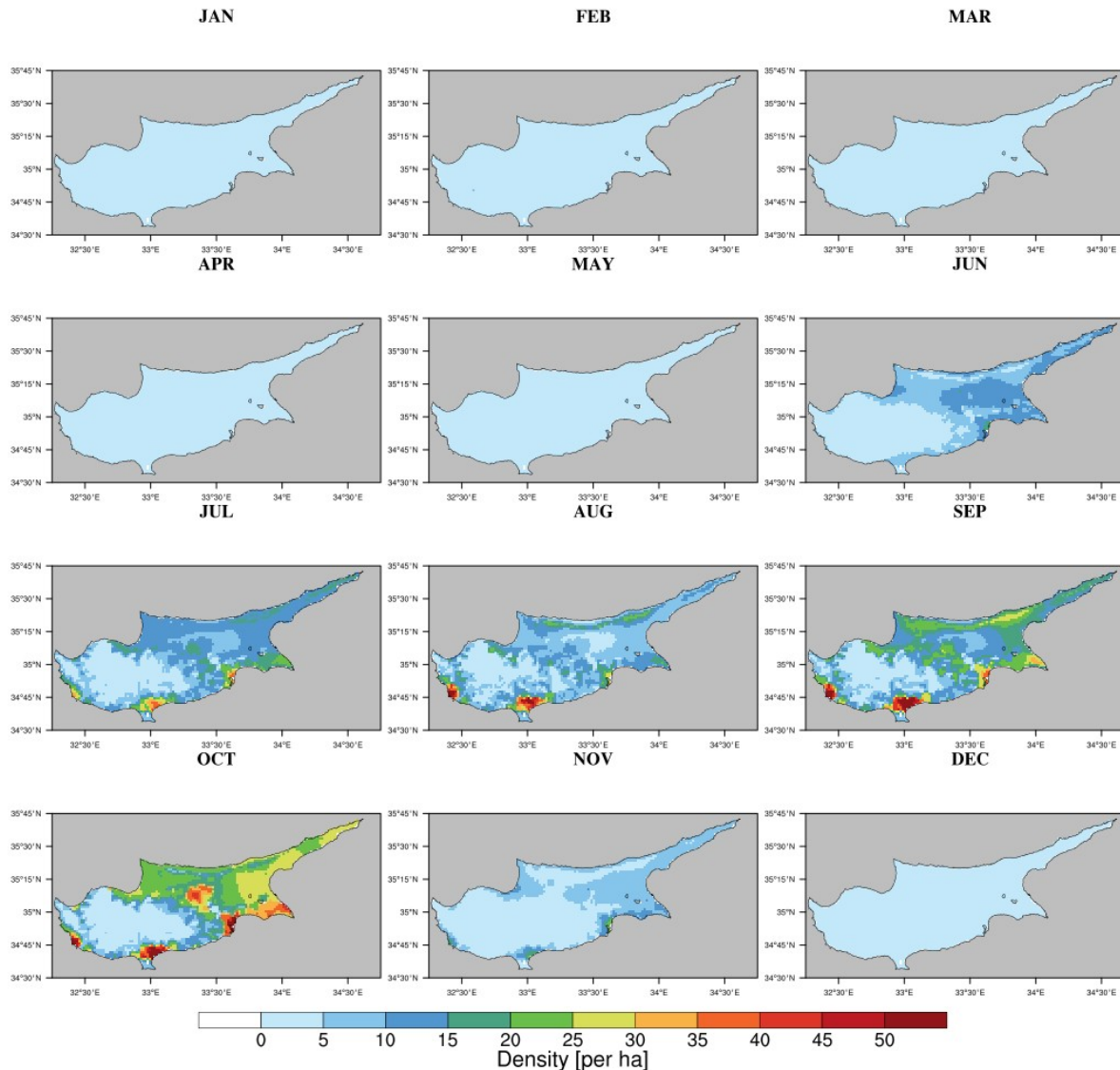


Top: Simulated adult *Ae. aegypti* density (per ha) by the VECTRI model for the periods 1990-2009 and 2010-22. Hatching denotes areas > 250 person/km². **Bottom:** Recent changes in simulated adult density (2010-22 vs 1990-2009).

Hotspots simulated over densely populated regions of Cyprus: Paphos, Limassol, Larnaca, Nicosia and Ayia Napa. Moderate risk simulated over the northern coasts.

Decrease in simulated abundance during the past 10-15y, excepting over Paphos.

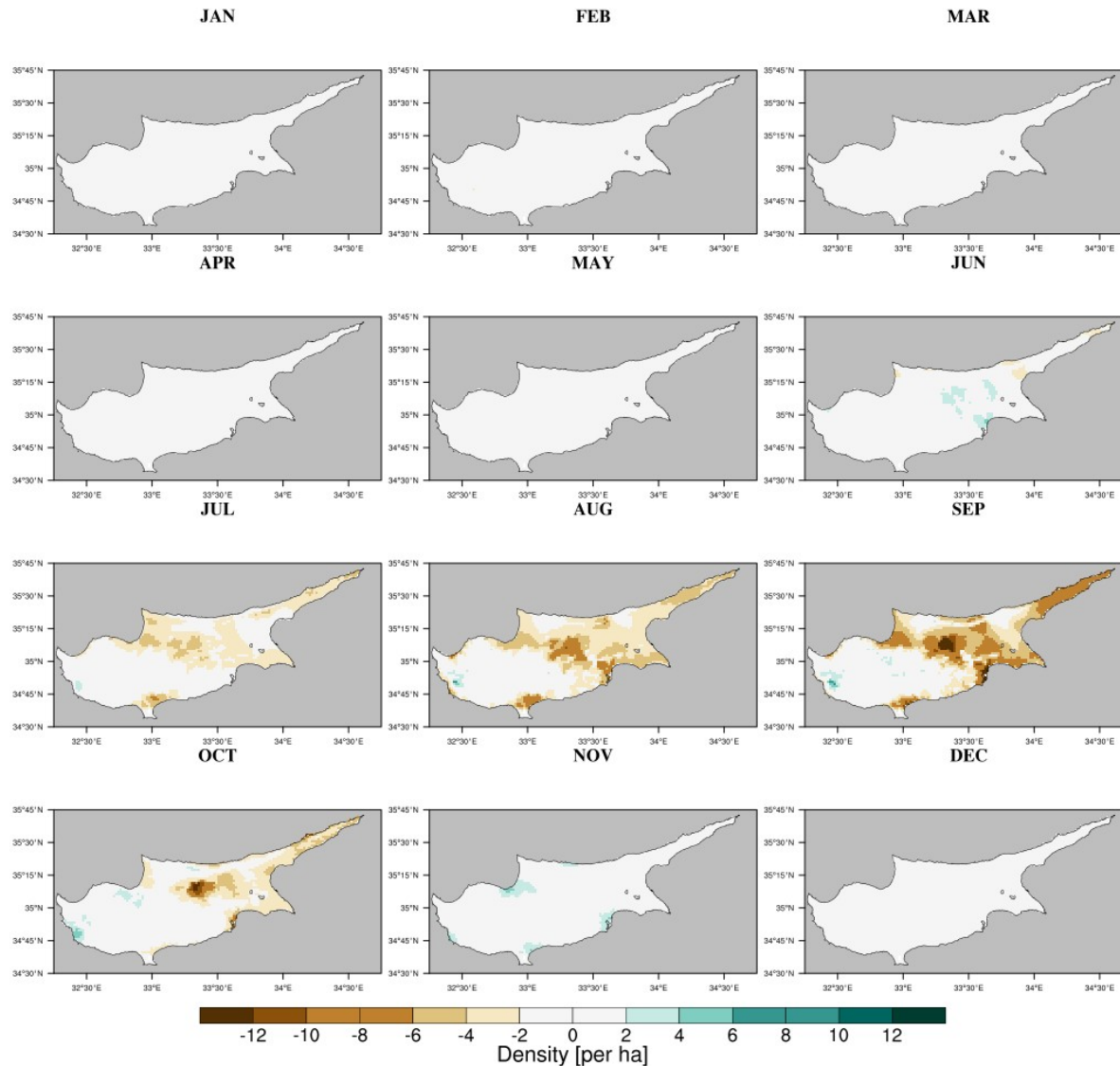
Simulated seasonal cycle in Cyprus: 1990-2022



Left: Mean seasonal cycle of simulated adult *Ae. aegypti* density (per ha) by the VECTRI model for 1990-2022.

Simulated mosquito activity between June and November with peaks in Sep-Oct over Cyprus.

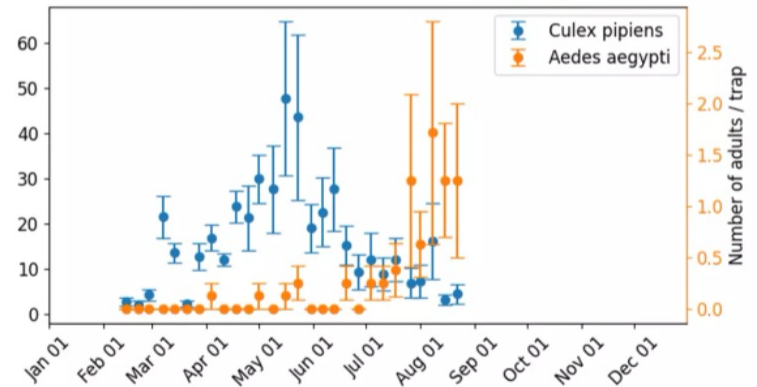
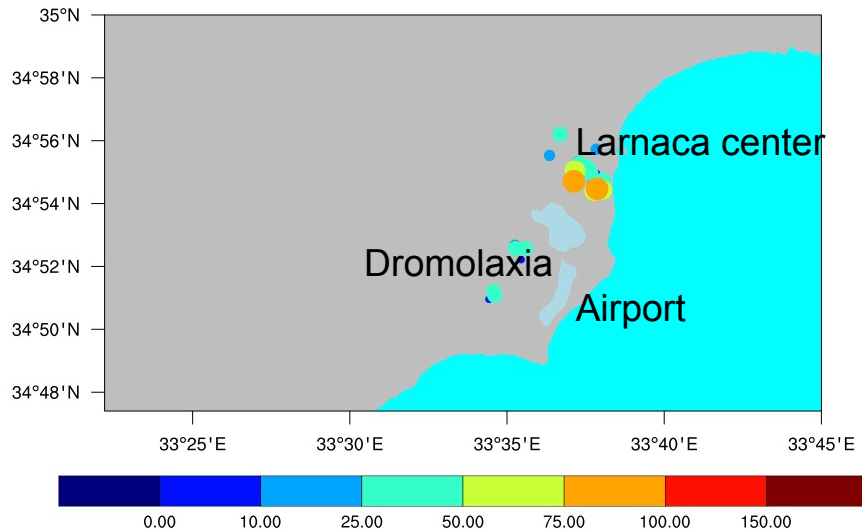
Simulated seasonal cycle in Cyprus: 2010-22 vs 1990-2009



Left: Changes in seasonal cycle for simulated *Ae. aegypti* density (per ha) between 2020-22 and 1990-2009.

Simulated decrease in abundance during the warmest months (Aug-Sep), in particular over Nicosia. Slight increase in climate suitability in June and November.

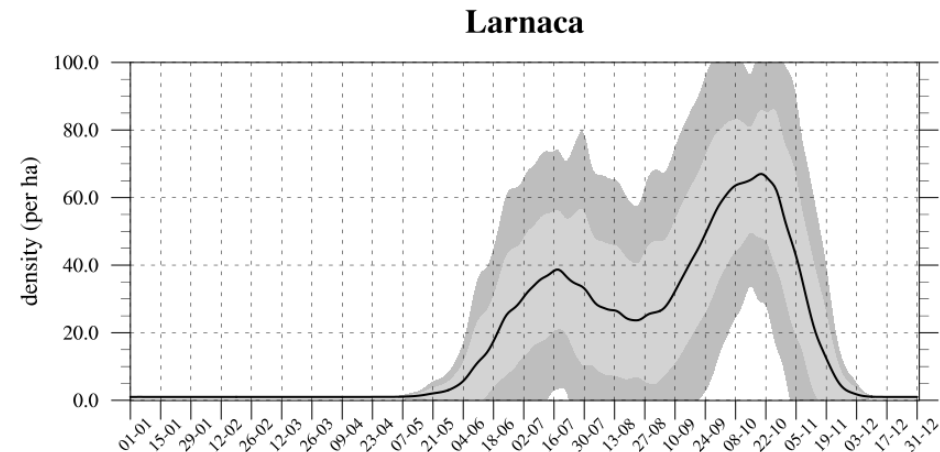
Ae. aegypti in Larnaca



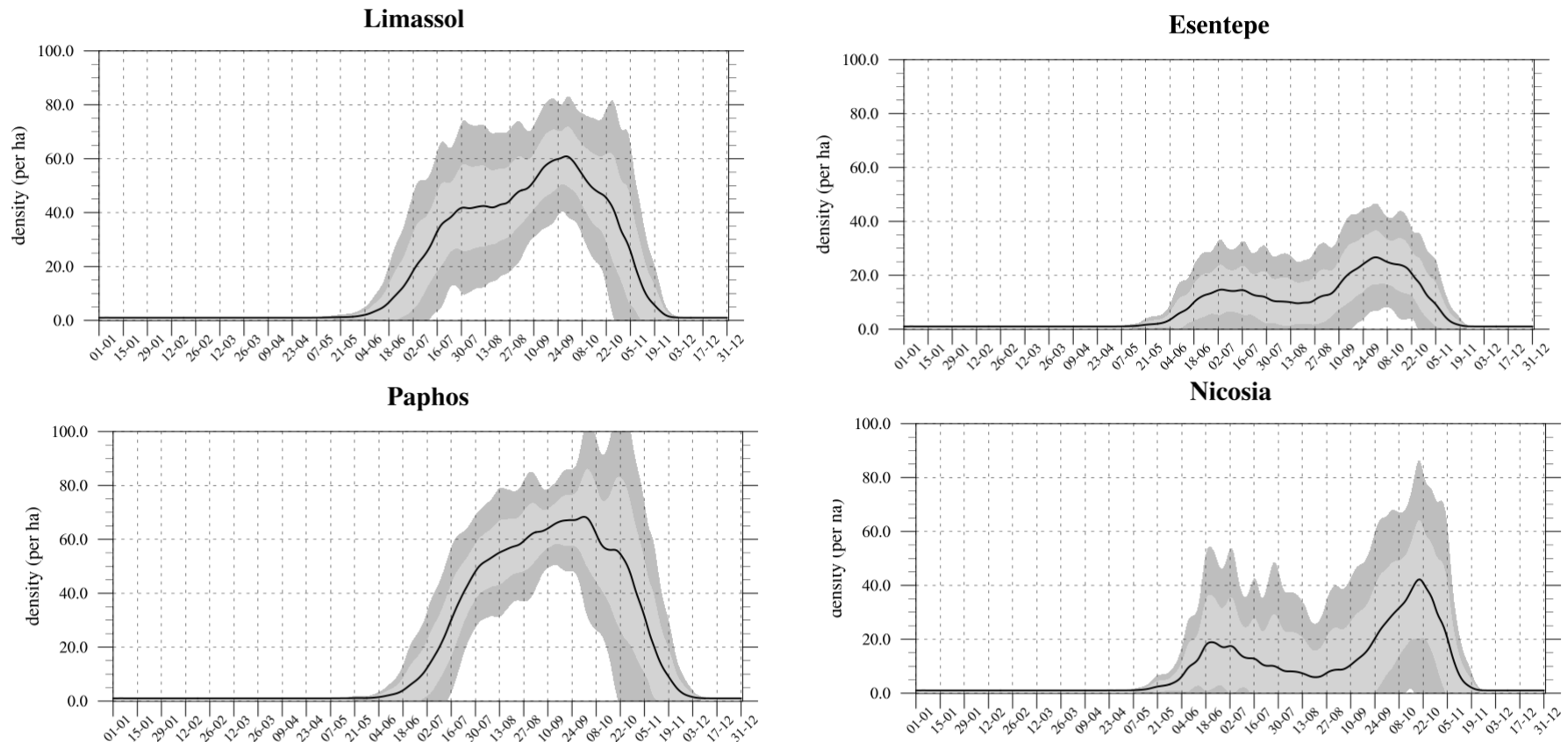
Top: Averaged egg abundance (per trap per week) for *Ae. aegypti* in Larnaca area for the period 24 Oct – 5 Dec 2022.

Top right: Number of adult *Ae. aegypti* in Larnaca (BGS trap data).

Bottom right: Simulated adult density (per ha) over Larnaca for the period 1990-2022.

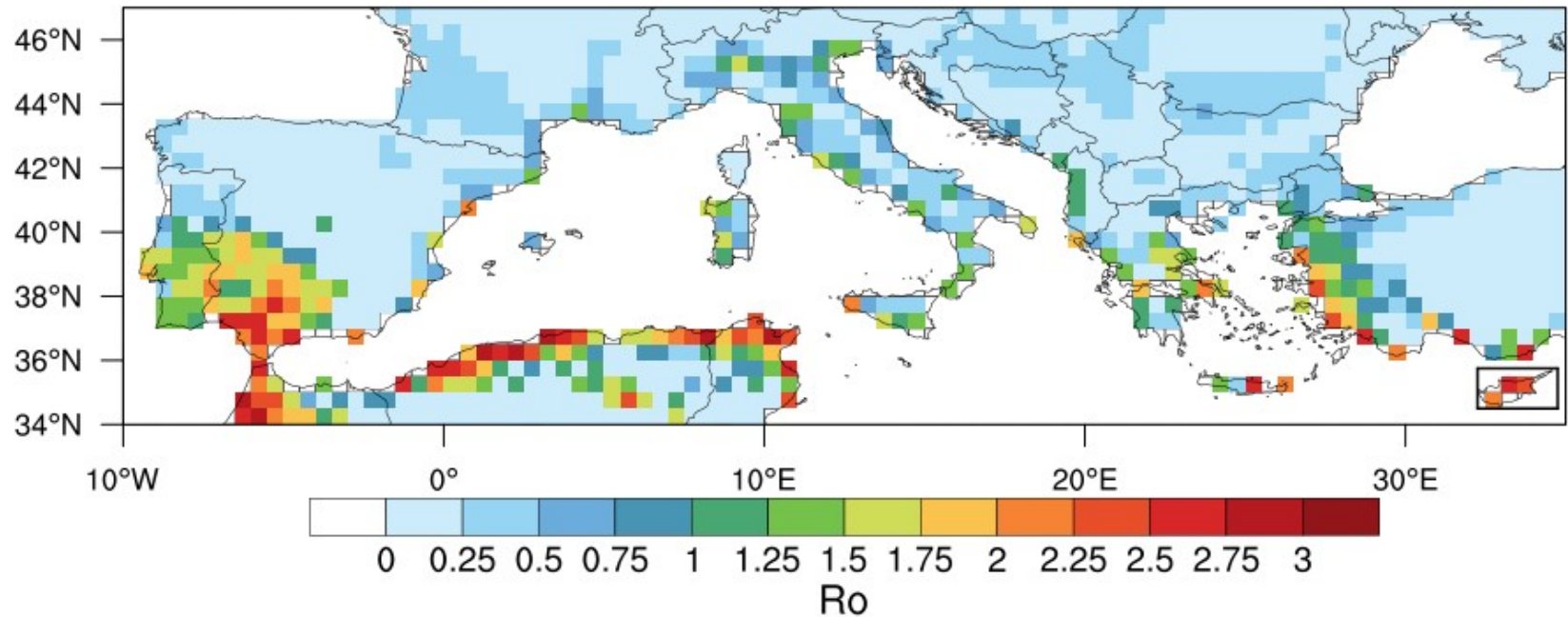


Simulated seasonal cycles of *Ae. aegypti* in cities



Mean seasonal cycle of simulated larval density for *Ae. aegypti* in Limassol, Esentepe, Paphos and Nicosia. The mean for 1990-2022 is depicted by the black solid line. The light-gray and gray envelope respectively show one and two standard deviations around the mean calculated for all years.

Risk of arbovirus transmission over the MED region



Annual mean basic reproduction number (R_0) for the risk of arbovirus transmission by *Ae. aegypti* & *Ae. albopictus* over the Mediterranean region (1980-2010). Cyprus is highlighted by the black box. One host – two vectors (*Ae. albopictus* and *Ae. aegypti*) R_0 model developed for ZIKV (Caminade et al., PNAS 2017).

These hotspots are consistent with **historical dengue epidemics** reported in **Cyprus** in 1861, 1888-89, 1913 and 1928; in **Crete** in 1881; over **western Greece** in 1889, 1895-97, 1910, 1927-28, 1929-33 ; in **Napoli** in 1889-90, in **Turkey** in 1889-90, 1899, 1916, 1927-28 and 1945 and over **Andalucía** in **Spain** in 1784-88-93, 1863-67, 1887 and 1928 (Schaffner & Mathis, Lancet Inf. Dis. 2014).

Largest documented dengue epidemic in Athens in 1927-28 (Louis C., Emerg Infect Dis. 2012).

Summary

- Statistical and dynamical models reproduce the historical distribution of *Ae. aegypti* over Europe. *Ae. aegypti* was present over the southern coasts of the Mediterranean, along the coasts of the Adriatic, and northern Africa.
- The VECTRI models simulate “climatic hotspots” over the urban areas of Larnaca, Limassol, Paphos, Ayia Napa, Nicosia and to a lesser extent the northern coasts. Recent climatic conditions less suitable over urban areas due to high temperature conditions over summer.
- Bimodal activity of *Ae. aegypti* for Larnaca, Nicosia and Esentepe (with a peak in June-July then October). Recent climate change tends to increase simulated mosquito density over the season edges and decrease during the warmest months in summer.
- Historical transmission of dengue and Yellow Fever occurred in main ports of the Mediterranean before vector/malaria control measures were undertaken post WWII.
- But *Ae. aegypti* has failed to re-establish in southern Europe. Other reasons might prevent its re-establishment (competition with other mosquito species? piped water? Lower potential to overwinter? Transportation by boat and/or aircraft?). See *Wint et al. (2022)*

Recommendations

- Stringent and continuous surveillance effort (deployment of ovitrap and BGS) should be developed and intensified in Larnaca but also Limassol, Paphos, Ayia Napa, Nicosia, and over the northern coast and eastern part of Cyprus (work in progress).
- Stringent surveillance of imported dengue, Zika and chikungunya cases in spring-summer in portal cities and at Larnaca international airport should be conducted to avoid potential autochthonous disease cases occurring during early summer and fall.
- Published epidemiological study found that “the introduction of dengue cases to Madeira in 2012 occurred around a month before first autochthonous cases were reported, during the maximum influx of airline travellers, and that declining temperature conditions in autumn were critical in limiting the lifespan of the dengue outbreak in early December 2012” (Lourenco et al., 2014) -> Learn from past epidemics
- Given that Greece is the main trading partner of Food products into Cyprus (World Bank, 2021), stringent surveillance should be conducted in containers transiting between the main harbours of both countries and in large airports.
- Sterile Insect Technique and other vector control approaches to eradicate *Ae. aegypti* in Cyprus.
- Model comparison exercise required to estimate uncertainties and provide science-informed decision making.

Acknowledgments

- Marlen I. Vasquez, Herodotos Herodotou, Gregoris Notarides, Costas Pavlou, Filip Fayad, Apostolos Papakonstantinou, Dusan Petric, Marios Violaris, Mamai Wadaka, Adrian M. Tompkins & Jeremy Bouyer.
- ICTP team, Italy: Adrian Tompkins & Miguel Garrido Zornoza.
- Universidad de Los Andes team in Colombia: Juan Daniel Umana Caro & Mauricio Santos Vega who are working on the *Ae. aegypti* version of VECTRI and dengue transmission risk in Colombia.
- Daniele Da Re, Kamil Erguler and the Climate-sensitive Vector Dynamics group.
- ICTP-IAEA project “Using the VECTRI mosquito model with machine learning to derive optimal strategies for the Sterile Insect Technique (SIT)”. National IAEA TC project CYP5020 “Developing a national rapid response strategy for the prevention of the establishment of the Asian tiger mosquito”. The delimitation strategy was supported by health workers of the MPHS and CUT researchers. The National Surveillance program was supported by CUT researchers.

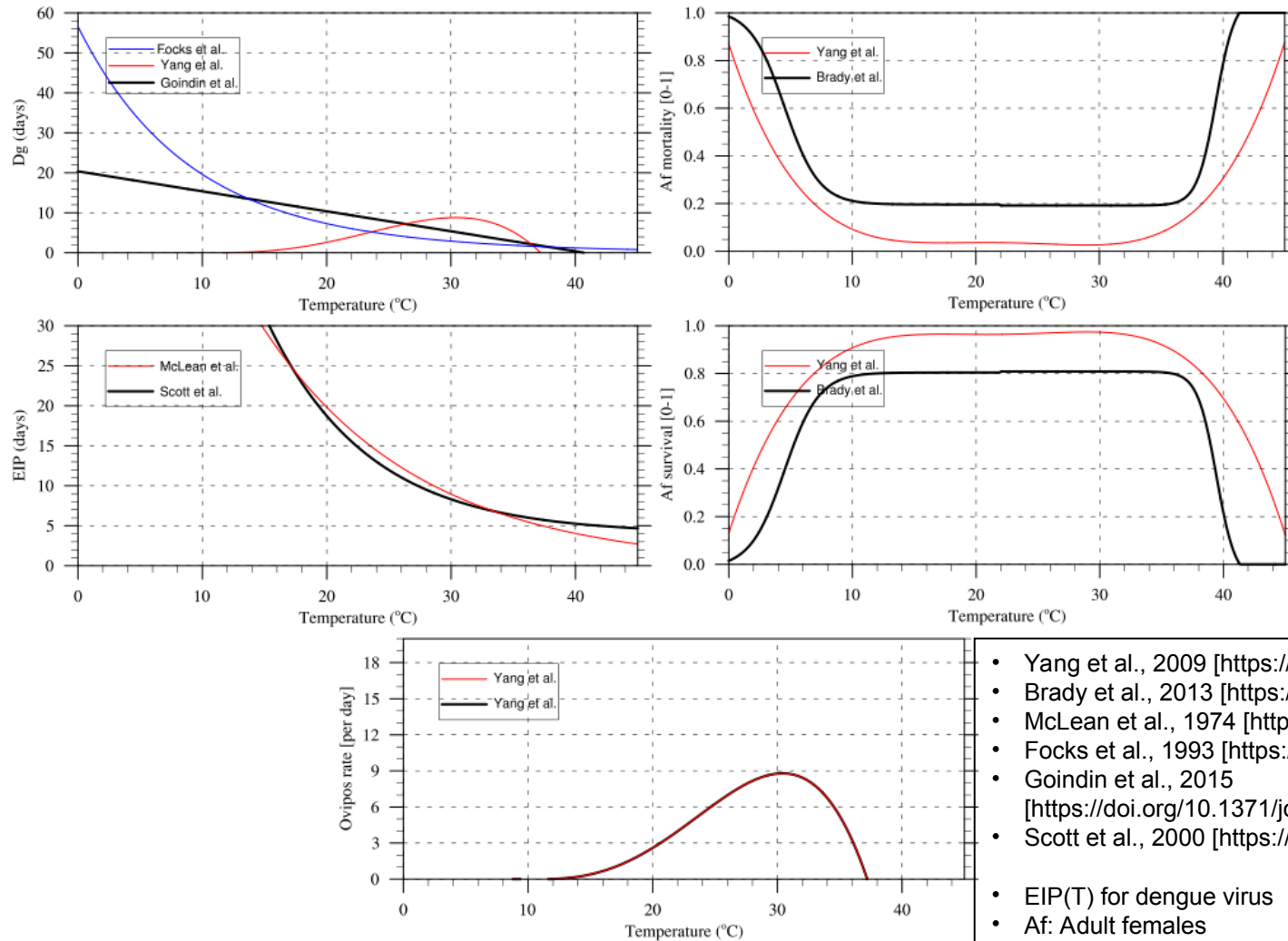


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Extra slides

Data & Methods



- Yang et al., 2009 [<https://doi.org/10.1017/S0950268809002040>]
- Brady et al., 2013 [<https://doi.org/10.1186/1756-3305-6-351>]
- McLean et al., 1974 [<https://doi.org/10.1139/m74-040>]
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- EIP(T) for dengue virus
- Af: Adult females

Ae. aegypti in Larnaca - animation

